spendwise

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SpendWise

Project Title

SpendWise – A Simple Backend Learning Project using Node.js, Express, MongoDB, and JWT

Project Overview

SpendWise is a backend application designed to help beginner developers learn how to build secure REST APIs using Node.js, Express, MongoDB, and JWT authentication. The project focuses on implementing user authentication and basic CRUD operations in a clean and simple way so students can understand how backend systems work in real-world scenarios.

Key Objectives:

- Provide secure user authentication using JWT
- Enable basic CRUD operations on user data (transactions/records)
- Demonstrate MongoDB integration using Mongoose
- Show how protected routes work using middleware
- Maintain a simple and beginner-friendly backend architecture

Target Users:

- Beginner backend developers
- Students learning Node.js and Express
- Freshers preparing for technical interviews
- Anyone who wants a simple project to understand JWT and MongoDB
- Learners who want hands-on practice with REST APIs

Project Instruction

SpendWise is a backend-only project built using Node.js and Express.

Quick Start:

npm init

npm install

npm run dev

Steps to Create the Project:

- Choose or create a directory for the backend project.
- Open terminal and navigate to that directory.
- Initialize Node.js project using npm init.

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- Install required packages: express, mongoose, jsonwebtoken, bcrypt, cors, dotenv.
- Create folders: models, routes, controllers, middleware, config.
- Start the server using npm run dev.
- Test APIs using Postman or Thunder Client at http://localhost:5000.

Pre-requisites:

Software Requirements

Software	**Purpose**
Node.js	JavaScript runtime for backend
MongoDB	NoSQL database for data storage
npm	Package manager
Git	Version control system
Code Editor	VS Code or similar

Knowledge Requirements

- JavaScript ES6+
- Node.js and Express.js
- MongoDB and Mongoose
- JWT authentication

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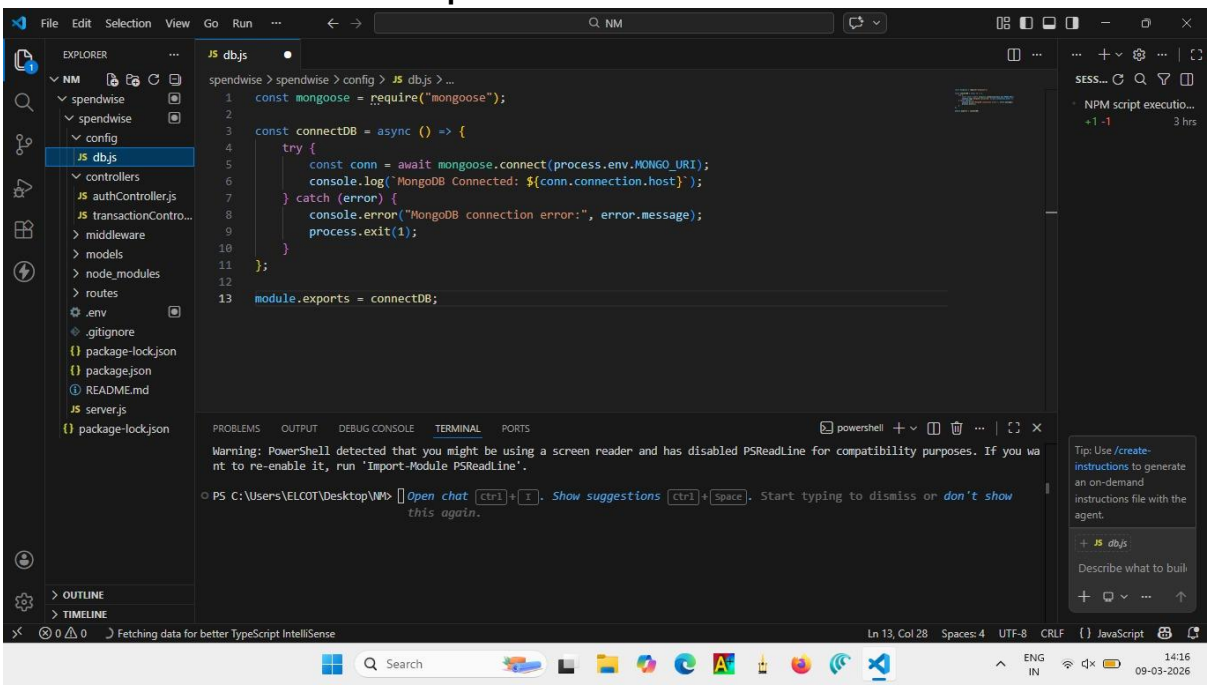
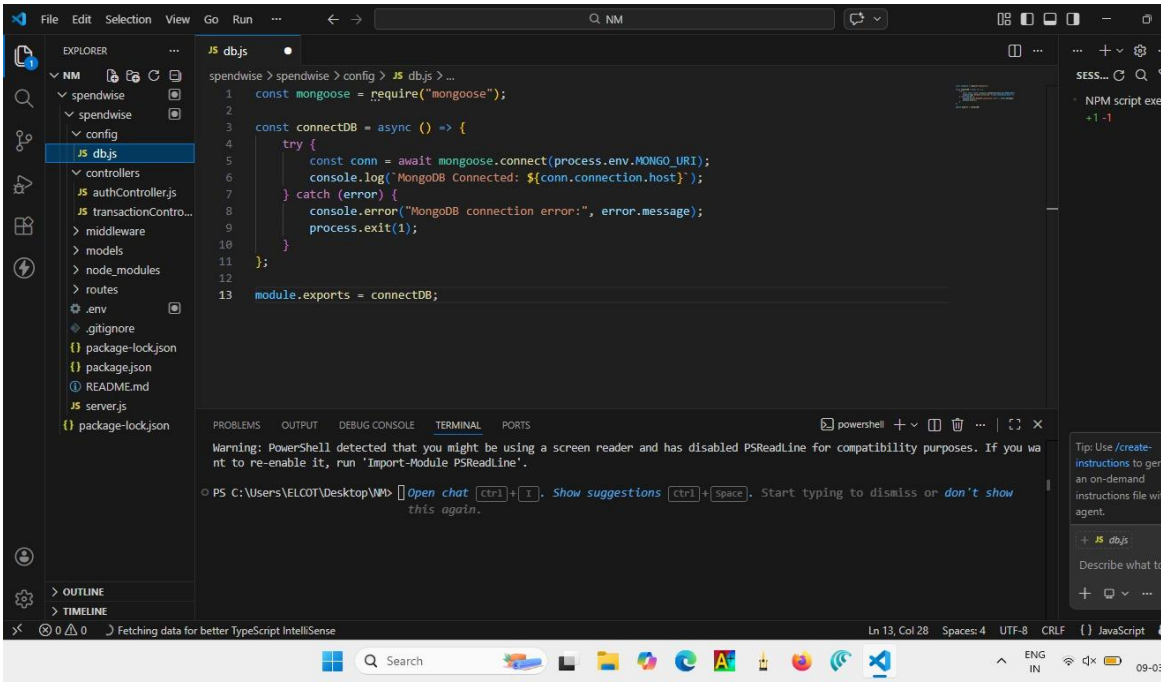


- Basic REST API concepts
- Git basics

Thumbnail for Project

- A simple backend project thumbnail representing:

Node.js + Express + MongoDB + JWT + CRUD APIs



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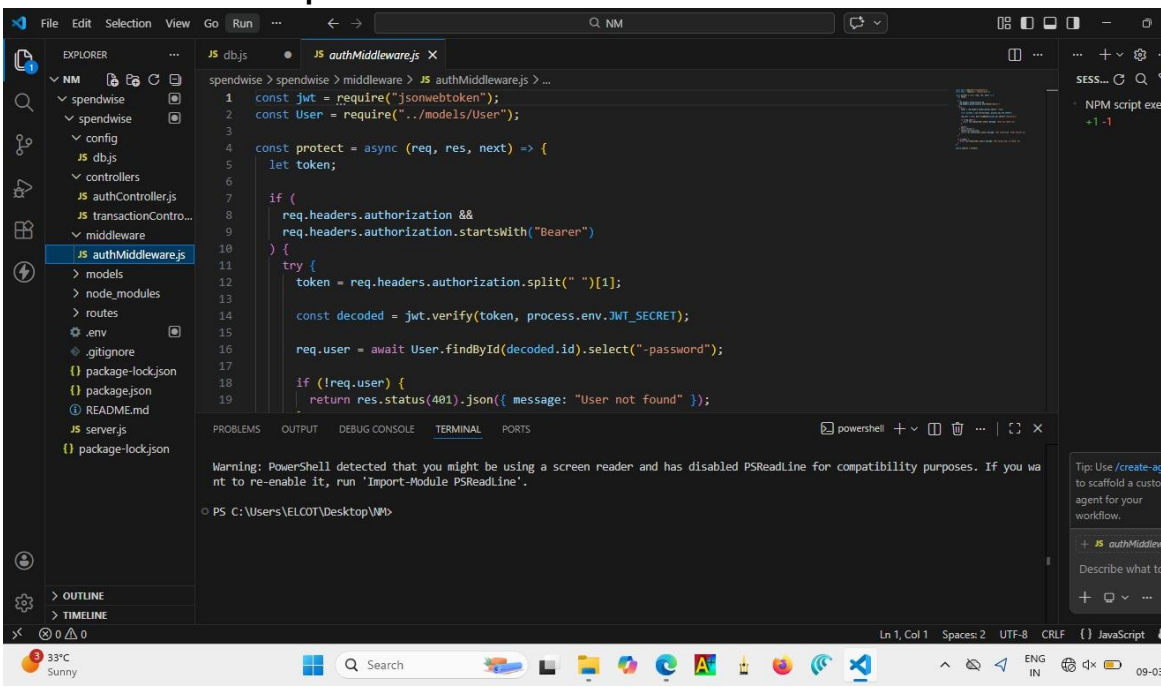
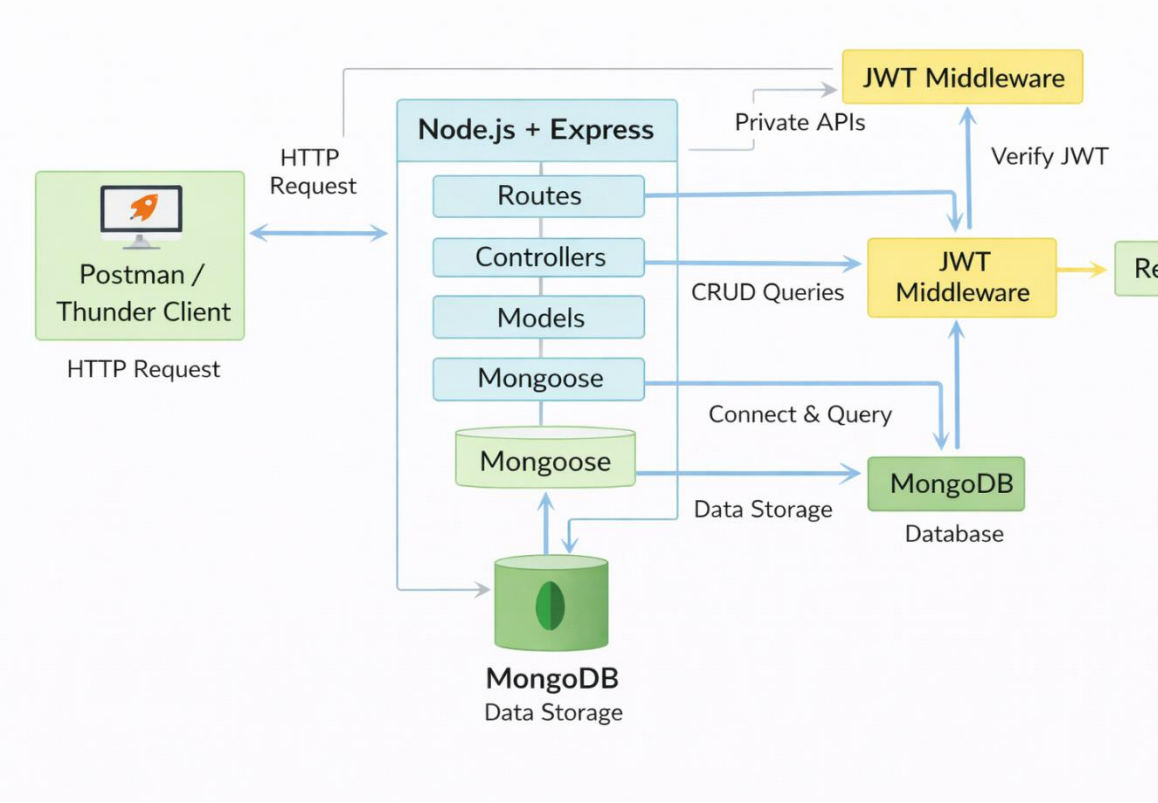


Project Architecture Image



Technical Architecture

Backend Architecture (Node.js + Express)

- **Controllers:** Handle business logic for authentication and CRUD

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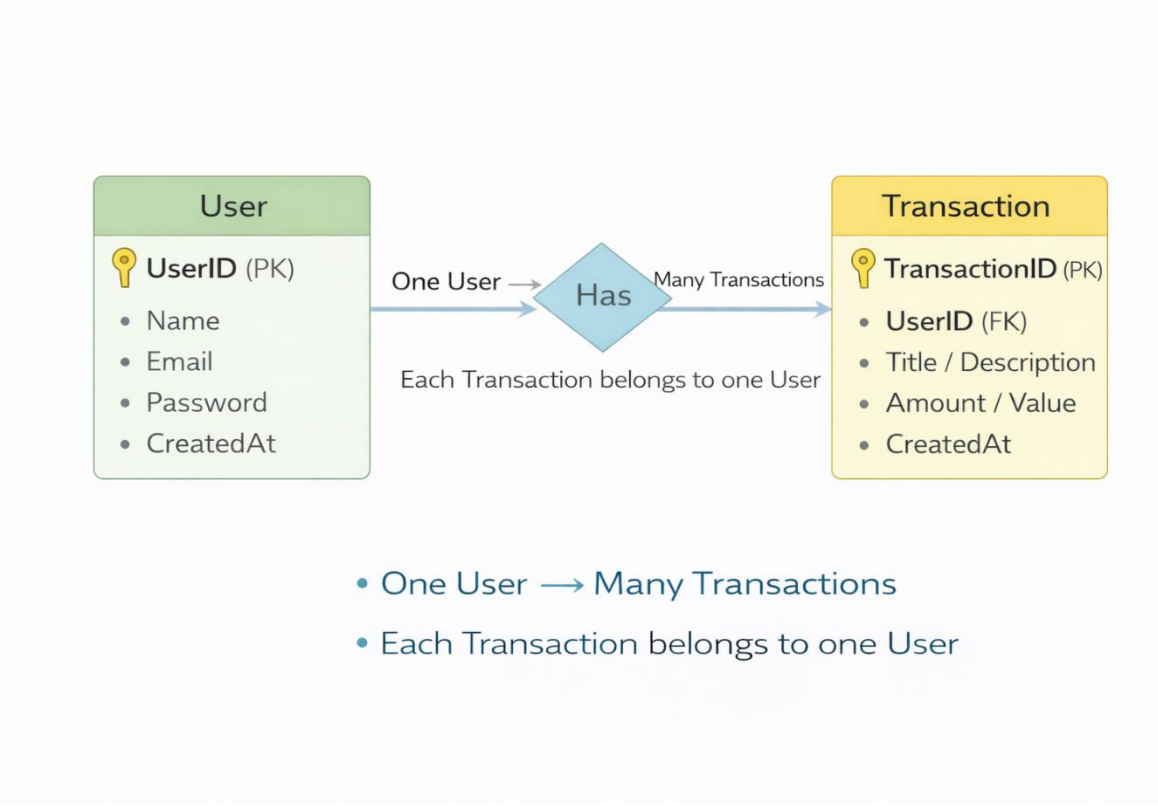
- **Models:** Mongoose schemas for User and Transaction
- **Routes:** API endpoints for auth and CRUD operations
- **Middleware:** JWT verification and request protection
- **Configuration:** Database connection and environment variables

Database Architecture (MongoDB)

- Users Collection:
  _id, name, email, password (hashed), createdAt
- Transactions Collection:
  _id, userId (FK), title/description, amount/value, createdAt
- Relationship:
  Many Transactions belong to One User

ER Diagram



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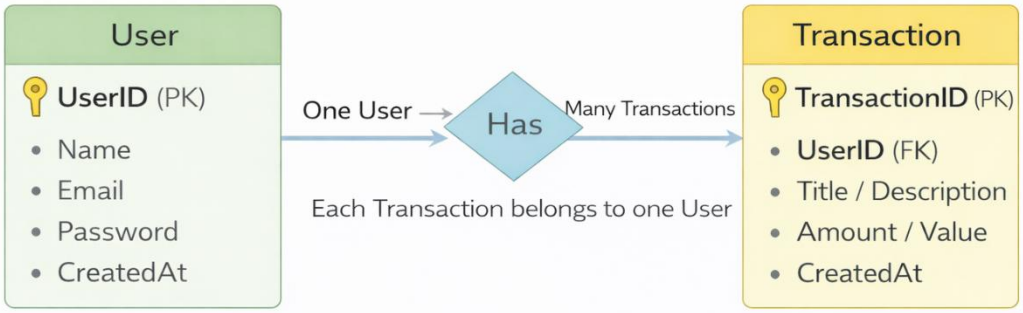

- One User ⟶ Many Transactions
- Each Transaction belongs to one User

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